

An MPC-based Pose Control Framework of Multi-Section Soft Manipulator using Hybrid Jacobian Estimation

Shuangquan Zou, Yueyong Lyu, Tanghao Qin, Guangfu Ma, Yanning Guo, and Chuanjiang Li

Abstract— Soft robotic manipulation is currently one of the most active problems in robotics, but simultaneously controlling the position and orientation of the end remains an open research problem for the soft manipulator. In this paper, a novel pose control framework based on model prediction control (MPC) is presented for the multi-section soft manipulator. Firstly, the continuum kinematics is established based on the piecewise constant curvature (PCC) model. Secondly, a hybrid method is proposed to estimate the Jacobian matrix in real time, which combines analytical and numerical methods. Then, an exponentially weighted MPC controller is developed to regulate the end pose in closed loop. Finally, simulations are carried out to verify the feasibility and effectiveness of the Jacobian matrix estimation method and pose control. The simulation results indicate that our controller performs better than peer works.

I. INTRODUCTION

As an emerging area of robotics, soft manipulators have been proven to be of unprecedented dexterity but challenging research. Unlike rigid manipulator, soft manipulators, made of flexible materials and structures, perform more flexibly in task that require compliant contact [1]. Therefore, it has great application potential in some scenarios with high safety requirements, such as medical treatment [2] and rescue [3].

So far, the majority of the work is focusing on structural [4] and prototype [5] design, and have made great progress. But the practical application of soft manipulators is still quite limited owing to the relatively lagging research progress of control. In a wide range of robotic applications, the pose control of the end actuator is the most basic control issue.

To avoid complex nonlinear modeling, some researchers combine the deep learning to explore the pose control of soft manipulators. Satheeshbabu et al. [6] proposed a deep reinforcement learning framework to establish the nonlinear characteristics between the actuation and the task space to complete the open-loop control. In the subsequent research [7], the state feedback is added to form the closed-loop control, which significantly improves the position error. Thuruthel et al. [8] constructed a deep neural network to learn the kinematics controller. In [9], the recurrent neural network and reinforcement learning are combined to realize closed-loop dynamic control. However, these learning methods require high-intensity experiments to generate databases, which not only takes a lot of time but is also a challenge for the self-made prototype. In addition, the obtained learning model is only applicable to the trained prototype.

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Therefore, model-based control is still the mainstream of the pose control scheme. Qi et al. [10] proposed a fuzzy kinematic controller with the membership function to overcome the uncertainty of the model. A controller based on the fuzzy adaptive and active disturbance rejection is designed to control the stiffness and position of the soft manipulator in [11]. Frazelle et al. [12] introduced a nonlinear control strategy for position control and proved its Lyapunov stability. Patterson et al. [13] developed a robust feedback controller to achieve position tracking with coupled dynamics and actuator constraints. And the most commonly used inverse kinematics controller [14] can locate and grab in various applications. All of the above studies do not involve the end orientation control, i.e., only the position. Still, orientation control is also crucial in some scenarios, such as minimally invasive surgery.

Since the single-section soft manipulator has 3 DOFs, it only needs to control the end position, but for multi-section, the end orientation is worth considering. Mahl et al. [15] utilized the first-order resolved rate controller to track the desired position and orientation of x and y . Lai et al. [16] adopted a P controller to enable the precise positioning while keeping the tip vertical, which can be applied to blood suction automation. In [17], the tip orientation is set as a constraint condition, and the inverse solution of the controller is formulated as a damped least-square optimization problem. More kinds of controllers need to be developed for the pose control of the multi-section soft manipulator. Model predictive control (MPC) provides a feasible solution because it computes control input by optimizing a cost function in a finite time interval [18] to satisfy other constraints while performing control tasks.

The main contribution of this work is focused on: 1) An MPC-based motion controller for position and orientation control of the multi-section soft manipulator. 2) A novel hybrid Jacobian matrix estimation method combines the advantages of analytical and numerical methods to avoid kinematic singularities while improving time cost and estimation accuracy. 3) A better performance compared to other control algorithms through simulation.

II. CONTINUUM KINEMATICS MODELING

A soft manipulator with three pneumatic muscle actuators (PMA) for one section is selected as prototype in this paper (Fig.1(a)), which was our previous work [19]. Noted that all proposed method in this paper is not limited to this kind of prototype. Piecewise constant curvature (PCC) provides the most widely used and typical approach for developing the kinematics of continuous soft manipulators due to its simplicity and rapidity. These assumptions need to be obeyed in the PCC modeling. 1) For any section, the deformation of the backbone behaves as a CC, and the actuator chambers are

parallel. 2) Adjacent backbone curves are tangent at the junction. 3) Any other strain is not considered.

A. Forward kinematics

Forward kinematics can be decomposed into two sub-maps for geometric derivation, as shown in Fig.1 (b). In actuator space, $\mathbf{q}_i = [q_{i1}, q_{i2}, q_{i3}]^T \in \mathbb{R}^3$ denotes the joint vector for the i th section, and q_{ij} is the length of the j th actuator, where $j \in \{1, 2, 3\}$ and $q_{ij} \in [q_{\min}, q_{\max}]$. Also, $\mathbf{s}_i = [l_i, \kappa_i, \phi_i]^T \in \mathbb{R}^3$ describes the arc vector in the configuration space, where $l_i, \kappa_i \in (0, \infty)$ and $\phi_i \in [-\pi, \pi]$ are the arc length, curvature and plane rotation angle of the, respectively. The closed-form solution can be calculated as

$$l_i(\mathbf{q}_i) = (q_{i1} + q_{i2} + q_{i3})/3, \quad (1)$$

$$\kappa_i(\mathbf{q}_i) = \frac{2\sqrt{q_{i1}^2 + q_{i2}^2 + q_{i3}^2 - q_{i1}q_{i2} - q_{i1}q_{i3} - q_{i2}q_{i3}}}{d(q_{i1} + q_{i2} + q_{i3})}, \quad (2)$$

$$\phi_i(\mathbf{q}_i) = \text{atan2}(\sqrt{3}(q_{i2} + q_{i3} - 2q_{i1})/3(q_{i2} - q_{i3})). \quad (3)$$

where d is the distance from the center of the baseplate to the center of the actuator. Then, the second mapping can solve the local end position of i th section in the task space $\mathbf{p}_i \in \mathbb{R}^3$ using $\mathbf{s}_i$ from the configuration space:

$$\begin{aligned} \mathbf{p}_i(\mathbf{s}_i) &= [p_i^x, p_i^y, p_i^z]^T \\ &= 1/\kappa_i [c\phi_i(1-c\theta_i), s\phi_i(1-c\theta_i), s\theta_i]^T \end{aligned} \quad (4)$$

where $c \triangleq \cos$ and $s \triangleq \sin$. θ_i is the bending angle of the backbone curve obtained from $\theta_i = l_i \kappa_i$. As for the kinematics of the multi-section soft manipulator, the coordinate transformation between adjacent sections can be derived by the homogeneous transformation matrix (HTM). The HTM for the i th section $\mathbf{T}_i$ is as follows

$$\mathbf{T}_i(\mathbf{s}_i) = \begin{bmatrix} \text{Rot} & \text{Trans} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_i(\mathbf{s}_i) & \mathbf{p}_i(\mathbf{s}_i) \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \quad (5)$$

Here,

$$\mathbf{R}_i(\mathbf{s}_i) = \begin{bmatrix} c^2\phi_i(c\theta_i-1)+1 & s\phi_i c\phi_i(c\theta_i-1) & c\phi_i s\theta_i \\ s\phi_i c\phi_i(c\theta_i-1) & s^2\phi_i(c\theta_i-1)+1 & s\phi_i s\theta_i \\ -c\phi_i s\theta_i & -s\phi_i s\theta_i & c\theta_i \end{bmatrix}. \quad (6)$$

It is worth noting that under the condition of $q_{i1} = q_{i2} = q_{i3}$, the rotation matrix $\mathbf{R}_i$ will become the 3×3 identity matrix $\mathbf{E}_{3 \times 3}$. Once the local HTM of each section is obtained, the global HTM of the whole manipulator ${}^N\mathbf{T}$ ($N \geq i$ is the section number) relative to the base coordinate $\{O_1\}$ can be realized recursively with multiplication.

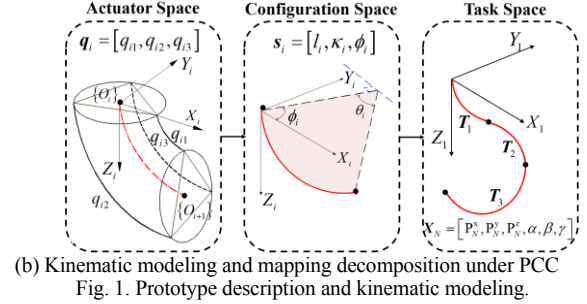
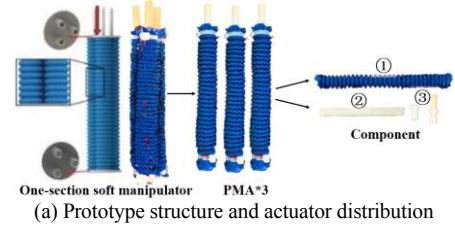
$${}^N\mathbf{T}(\mathbf{s}) = \prod_{i=1}^N \mathbf{T}_i(\mathbf{s}_i) = \begin{bmatrix} \Phi_N(\mathbf{s}) & \mathbf{P}_N(\mathbf{s}) \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \quad (7)$$

where the orientation matrix $\Phi_N \in \mathbb{R}^{3 \times 3}$ and end position vector $\mathbf{P}_N = [P_N^x, P_N^y, P_N^z]^T \in \mathbb{R}^3$ are defined as

$$\Phi_N(\mathbf{s}) = \prod_{i=1}^N \mathbf{R}_i(\mathbf{s}_i) \quad (8)$$

$$\mathbf{P}_N(\mathbf{s}) = \mathbf{P}_{N-1} + \Phi_{N-1} \mathbf{p}_N. \quad (9)$$

To sum up, the forward kinematics of the soft manipulator with N sections can be generated by combining (1)-(9).



B. Orientation kinematics

In the task space, not only the position of the end-effector $\mathbf{P}_N$ but also the orientation angle is a helpful performance of the entire control task. Φ_N is usually decomposed into three orientation components $\boldsymbol{\varphi} = [\alpha, \beta, \gamma]^T \in \mathbb{R}^3$ represented by Euler angles according to Z-Y-X rotation order.

$$\Phi_N(\mathbf{s}) = \mathbf{R}_z(\gamma) \mathbf{R}_y(\beta) \mathbf{R}_x(\alpha) \quad (10)$$

The Euler angles are computed with the elements of Φ_N .

$$\begin{aligned} \alpha &= \text{atan2}(\Phi_N^{32}, \Phi_N^{33}) \\ \beta &= \text{asin}(-\Phi_N^{31}) \\ \gamma &= \text{atan2}(\Phi_N^{21}, \Phi_N^{11}) \end{aligned} \quad (11)$$

Combined with position $\mathbf{P}_N$ and orientation $\boldsymbol{\varphi}$, the pose of the multi-section soft manipulator $\mathbf{X}_N \in \mathbb{R}^6$ is defined as

$$\mathbf{X}_N(\mathbf{s}) = [\mathbf{P}_N^T, \boldsymbol{\varphi}^T]^T = [P_N^x, P_N^y, P_N^z, \alpha, \beta, \gamma]^T \quad (12)$$

C. Velocity kinematics

It can be seen from the kinematics model that any small movements of the joints will cause slight changes in the pose of the end effector. This differential relationship with respect to time can be simply expressed by the Jacobian matrix as

$$\dot{\mathbf{X}} = \mathbf{J}(\mathbf{q}) \dot{\mathbf{q}} \quad (13)$$

Formula (13) can be written in more detail as

$$\begin{bmatrix} \mathbf{v} \\ \mathbf{w} \end{bmatrix} = \begin{bmatrix} \mathbf{J}^v(\mathbf{q}) \\ \mathbf{J}^w(\mathbf{q}) \end{bmatrix} \dot{\mathbf{q}} \quad (14)$$

where $\mathbf{v} = [v_x, v_y, v_z]^T \in \mathbb{R}^3$ and $\mathbf{w} = [w_x, w_y, w_z]^T \in \mathbb{R}^3$ are the linear and angular velocities of the end effector, respectively, and $\mathbf{J}^v \in \mathbb{R}^{3 \times 3N}$ and $\mathbf{J}^w \in \mathbb{R}^{3 \times 3N}$ are the corresponding linear velocity and angular velocity Jacobian.

The Jacobian matrix can be seen as a means of local linearization, so we assume that the soft manipulator moves at a low speed, and the pose changes continuously and gently.

III. JACOBIAN ESTIMATION METHOD

In this section, analytical and numerical methods for solving the Jacobian matrix of multi-section soft manipulator are proposed to eventually explore a new hybrid approach.

A. Analytical method

The most direct way to solve the Jacobian matrix is to develop the partial derivative of the established kinematic analytical model according to the Jacobian definition.

$$\mathbf{J}(\mathbf{q}) = \frac{\partial \mathbf{X}(\mathbf{s})}{\partial \mathbf{q}} = \frac{\partial \mathbf{X}(\mathbf{s})}{\partial \mathbf{s}} \frac{\partial \mathbf{s}}{\partial \mathbf{q}} \quad (15)$$

Firstly, the m -th column of the linear velocity Jacobian matrix considering N -section soft manipulator is depicted as

$$\{\mathbf{J}_N^v(\mathbf{q})\}_{m=3(i-1)+j} = \frac{\partial \mathbf{p}_N}{\partial q_{ij}} = \frac{\partial (\mathbf{p}_{N-1} + \Phi_{N-1} \mathbf{p}_N)}{\partial q_{ij}}. \quad (16)$$

Formula (16) is divided into two cases to discussed. When $q_{ij} \in \mathbf{q}_N$, It can be simplified as

$$\{\mathbf{J}_N^v(\mathbf{q})\}_m = \Phi_{N-1} + \frac{\partial \mathbf{p}_N}{\partial q_{ij}}. \quad (17)$$

When $q_{ij} \in \mathbf{q}^{N-1}$, formula (18) is substituted by

$$\{\mathbf{J}_N^v(\mathbf{q})\}_m = \frac{\partial \mathbf{p}_{N-1}}{\partial q_{ij}} + \frac{\partial \Phi_{N-1}}{\partial q_{ij}} \mathbf{p}_N = \{\mathbf{J}_{N-1}^v(\mathbf{q})\}_m + \frac{\partial \Phi_{N-1}}{\partial q_{ij}} \mathbf{p}_N. \quad (18)$$

In the same way, the analytical formula of the angular velocity Jacobian matrix can be deduced based on (14).

$$\mathbf{w}(\mathbf{q}) = \mathbf{J}^w(\mathbf{q}) \dot{\mathbf{q}} = [\dot{\Phi}_N \Phi_N^T]^{\times} \quad (19)$$

It is well-known that the relationship between angular velocity and rotation matrix is $[\mathbf{w}]_{\times} = \dot{\Phi}_N \Phi_N^T$ where the

$$\text{skew-symmetric matrix } [\mathbf{w}]_{\times} = \begin{bmatrix} 0 & -w_z & w_y \\ w_z & 0 & -w_x \\ -w_y & w_x & 0 \end{bmatrix} \text{ and } [\cdot]^{\times}$$

is defined as the inverse operator of the skew-symmetric.

The angular velocity Jacobian matrix of the N -section soft manipulator is yielded by

$$\mathbf{J}_N^w(\mathbf{q}) = \left[\frac{\partial \Phi_N}{\partial \mathbf{q}^N} \Phi_N^T \right]^{\times} \quad (20)$$

Similarly, when $q_{ij} \in \mathbf{q}_N$, the m -th column of (20) becomes

$$\{\mathbf{J}_N^w(\mathbf{q})\}_{m=3(i-1)+j} = \left[\Phi_{N-1} \frac{\partial \mathbf{R}_N}{\partial q_{ij}} \mathbf{R}_N^T \Phi_{N-1}^T \right]^{\times}. \quad (21)$$

And when $q_{ij} \in \mathbf{q}^{N-1}$, formula (20) results in

$$\{\mathbf{J}_N^w(\mathbf{q})\}_m = \left[\frac{\partial \Phi_{N-1}}{\partial q_{ij}} \Phi_{N-1}^T \right]^{\times} = \{\mathbf{J}_{N-1}^w(\mathbf{q})\}_m. \quad (22)$$

For the sake of space, the explicit analytical expression of the Jacobian matrix is not reported but can be implemented recursively according to (17)-(21).

B. Numerical method

Linear Kalman filter (LKF), a numerical iterative method, has a great approximation ability for the state variables of linear systems, only using the input and output data, which is suitable for the online estimation of the Jacobian matrix. Formula (13) is discretized into the following first-order form. For the ease of expression, we only mark the iteration step k of all variables and omit other relevant arguments.

$$\mathbf{X}(k) = \mathbf{X}(k-1) + \mathbf{J}(k) \Delta \mathbf{q}(k) \quad (23)$$

where $\Delta \mathbf{q}(k) = \mathbf{q}(k) - \mathbf{q}(k-1) \in \mathbb{R}^{3N}$, and we set the state variables of the system like

$$\mathbf{x}(k) = [\{\mathbf{J}(k)\}^1, \dots, \{\mathbf{J}(k)\}^6]^T \in \mathbb{R}^{18N \times 1} \quad (24)$$

where $\{\mathbf{J}(k)\}^n$ is labeled as the n -th row of $\mathbf{J}(k)$, and the output of the system $\mathbf{y}(k) = \mathbf{X}(k) - \mathbf{X}(k-1) \in \mathbb{R}^{6 \times 1}$. Then we can construct a linear stochastic system without control input:

$$\begin{cases} \mathbf{x}(k) = \mathbf{x}(k-1) + \boldsymbol{\eta} \\ \mathbf{y}(k) = \mathbf{C}(k) \mathbf{x}(k) + \boldsymbol{\rho} \end{cases} \quad (25)$$

where $\boldsymbol{\eta} \sim N(0, \mathbf{A})$ and $\boldsymbol{\rho} \sim N(0, \mathbf{B})$ are process noise and measurement noise respectively, and $\mathbf{A} \in \mathbb{R}^{18N \times 18N}$ and $\mathbf{B} \in \mathbb{R}^{6 \times 6}$ are corresponding covariance matrices. The diagonal matrix $\mathbf{C}(k) = \text{diag}(\underbrace{\Delta \mathbf{q}(k)^T, \dots, \Delta \mathbf{q}(k)^T}_6) \in \mathbb{R}^{6 \times 18N}$. The

calculation process of LKF includes two parts: prediction and update, namely:

$$\begin{aligned} \tilde{\mathbf{P}}(k) &= \mathbf{P}(k-1) + \mathbf{A} \\ \tilde{\mathbf{x}}(k) &= \mathbf{x}(k-1) \\ \mathbf{H}(k) &= \tilde{\mathbf{P}}(k) \mathbf{C}(k)^T [\mathbf{C}(k) \tilde{\mathbf{P}}(k) \mathbf{C}(k)^T + \mathbf{B}]^{-1} \\ \mathbf{x}(k) &= \tilde{\mathbf{x}}(k) + \mathbf{H}(k) (\mathbf{y}(k) - \mathbf{C}(k) \tilde{\mathbf{x}}(k)) \\ \mathbf{P}(k) &= (\mathbf{E}_{18N} - \mathbf{H}(k) \mathbf{C}(k)) \tilde{\mathbf{P}}(k) \end{aligned} \quad (26)$$

where $\tilde{\mathbf{x}}(k)$ and $\tilde{\mathbf{P}}(k) \in \mathbb{R}^{18N \times 18N}$ are noted as the predicted values of the system state and state covariance matrix at step k . Hence, given the LKF as expressed in (26), $\mathbf{x}(k)$ is updated based on $\tilde{\mathbf{x}}(k)$, $\tilde{\mathbf{P}}(k)$ and $\mathbf{y}(k)$, which realizes the online estimation of Jacobian matrix in the iterative process.

C. Hybrid method

Although both analytical and numerical methods can achieve the estimation of Jacobian matrix, they have their own limitations, as listed below.

The limitations of the analytical method are apparent. 1) High calculation cost: The analytical method can be regarded as the process of continuous partial derivation of task space to joint space based on PCC kinematics as (15), which requires some calculation time. When it involves additional iterative calculation between sections should be considered with (17)-(22), further leading to the high calculation cost. 2) Geometric singularities: When the soft manipulator is in a state of pure elongation, i.e., cases $q_{i1} = q_{i2} = q_{i3}$, ϕ_i becomes undefined (see (3)), which means that all formulas containing ϕ_i are invalid, and the Jacobian matrix cannot be estimated.

The numerical method has a fast calculation speed, and since it does not depend on the kinematic model, there is no

singularity problem. However, it also has some limitations. 1) Infeasible results: As a numerical iterative method, without physical constraints, it will produce some results that exceed mechanical capabilities. For example, the solved q_{ij} is too small or even negative in value, which is impossible for the prototype. 2) Crucial initial value: Numerical iteration methods usually require an initial value of the Jacobian matrix $\mathbf{J}(0)$, which determines the calculation direction of the iteration. Bad initial values will make the value of the Jacobian matrix iterate in the wrong direction and even eventually cause the control system to be unable to converge.

As a consequence, a hybrid method is presented to combine the advantages of these two methods in this paper. Firstly, $\mathbf{J}(0)$ is selected by the analytical method, and then it is iterated based on LKF to estimate the Jacobian matrix online (refer to (26)), providing fast and reliable Jacobian matrix input for the subsequent controller.

IV. CONTROLLER DESIGN

In this section, an MPC framework is designed to achieve the desired pose for a multi-section soft manipulator. The MPC-based algorithm calculates the optimal control input by solving the quadratic cost function in the prediction interval and rolling optimization, minimizing the pose error $\mathbf{e}(k) = \mathbf{X}^* - \mathbf{X}(k)$ between the actual pose $\mathbf{X}(k)$ and a desired pose $\mathbf{X}^*$. Referring to the formula (23), the predicted pose at step $k + g$ is satisfied below:

$$\mathbf{X}^p(k+g) = \mathbf{X}(k) + \mathbf{J}(k) \cdot \mathbf{u}(k+g) \quad (27)$$

where $g \in [0, G]$ represents the length of the prediction interval and the control input $\mathbf{u}(k+g) = \Delta \mathbf{q}(k+g)$. To ensure the pose of the soft manipulator changes smoothly, the reference pose during the motion process is planned as

$$\mathbf{X}^r(k+g) = \mathbf{X}^* - e^{-\rho g} \cdot \mathbf{e}(k) \quad (28)$$

where ρ is a positive constant. The error $\mathbf{e}(k+g)$ between the reference and the prediction pose at step $k + g$ is derived

$$\begin{aligned} \mathbf{e}(k+g) &= \mathbf{X}^r(k+g) - \mathbf{X}^p(k+g) \\ &= (1 - e^{-\rho g}) \mathbf{e}(k) - \mathbf{J}(k) \mathbf{u}(k+g) \end{aligned} \quad (29)$$

The quadratic cost function from k to $k + g$ is set as

$$U = \frac{1}{2} \sum_{g=0}^G \mu^g [\mathbf{e}(k+g)^T \mathbf{Q} \mathbf{e}(k+g)] + \mathbf{u}(k)^T \mathbf{W} \mathbf{u}(k) \quad (30)$$

where $0 < \mu \leq 1$, and μ^g is discount factor. Assume that $\mathbf{u}(k)$ remains constant within the prediction interval, $\mathbf{Q}$ is identity matrix, and $\mathbf{W}$ is symmetric and positive definite matrices. Minimize U from k to $k + G$ to solve $\mathbf{u}(k)$. Combined with formula (29) and (30), the optimization process can be expressed as

$$\min \frac{1}{2} \left(\sum_{g=0}^G \mu^g \left\| (1 - e^{-\rho g}) \mathbf{e}(k) - g \mathbf{J}(k) \mathbf{u}(k) \right\|_2^2 + \mathbf{u}(k)^T \mathbf{W} \mathbf{u}(k) \right). \quad (31)$$

The first part of (31) guarantees that the error is minimized, while the second part is to minimize the $\mathbf{u}(k)$ so that the pose of the soft manipulator changes slowly. To

minimize U , take the derivative of (31) with respect to $\mathbf{u}(k)$ and set the gradient $\nabla = 0$ as

$$\nabla = \sum_{g=0}^G -g \mu^g \mathbf{J}(k)^T \left((1 - e^{-\rho g}) \mathbf{e}(k) - g \mathbf{J}(k) \mathbf{u}(k) \right) + \mathbf{W} \mathbf{u}(k) \quad (32)$$

Then the analytical formula of $\mathbf{u}(k)$ can be determined as

$$\mathbf{u}(k) = (a \mathbf{J}(k) + \mathbf{J}(k)^T \mathbf{W})^+ (b - c) \mathbf{e}(k) \quad (33)$$

where $a = (\mu^G - 2b) / \ln \mu$, $b = (G \mu^G \ln \mu - \mu^G + 1) / \ln^2 \mu$, and $c = (G(\mu e^{-\rho})^G \ln(\mu e^{-\rho}) - (\mu e^{-\rho})^G + 1) / \ln^2(\mu e^{-\rho})$. $\mathbf{J}^+$ is acquired by $\mathbf{J}^+ = \mathbf{J}^T (\mathbf{J} \mathbf{J}^T)^{-1}$. Finally, the stability of the control system is proved. Substitute (33) into (23) to get

$$(a \mathbf{E}_{6 \times 6} + \mathbf{J}(k)^T \mathbf{W} \mathbf{J}(k)^+)(\mathbf{e}(k) - \mathbf{e}(k-1)) + (b - c) \mathbf{e}(k) = 0. \quad (34)$$

Since $a > 0$, $b - c > 0$, and $\mathbf{J}(k)^T \mathbf{W} \mathbf{J}(k)^+$ is a positive definite matrix, it is evident that the error converges to 0, and the system is stable. The block diagram of the closed-loop control system proposed in this paper is illustrated in Fig. 2.

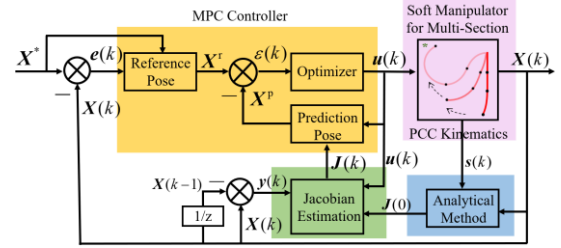


Fig. 2 Framework of the real-time pose control of the soft robot with multi-section.

V. SIMULATION RESULTS

In this section, the proposed Jacobian matrix estimation and MPC-based control design are verified by simulation. All simulations are considered a three-section soft manipulator displayed in Fig. 1, and all variables share the unit of mm except for the orientation in rad. The initial joint vector $\mathbf{q}(0) = [89, 89, 90, 90, 90, 89, 89, 90, 89]^T$, which is more closely to the natural elongation state of the prototype. In view of the mechanical performance of the prototype, the bounded constraint of the joint is added as $70 \leq q_{ij} \leq 300$.

A. Validation of the Jacobian Estimation

The advantages and limitations of the three proposed Jacobian estimation methods are validated through the following numerical cases. The error criterion is introduced to quantify the performance of Jacobian estimation

$$T_1 = \left\| \Delta \mathbf{X}(k) - \hat{\mathbf{J}}(k) \Delta \mathbf{q} \right\|_2^2. \quad (35)$$

where $\Delta \hat{\mathbf{X}}(k) = \hat{\mathbf{J}}(k) \Delta \mathbf{q}(k)$ is the estimated pose increment calculated by the estimated Jacobian matrix $\hat{\mathbf{J}}$, while $\Delta \mathbf{X}(k)$ is the real value calculated by kinematics.

Firstly, iterate 100 steps with the fixed $\Delta \mathbf{q} = [1, 1, 2, 1, 2, 1, 2, 1, 1]^T$, and monitor the T_1 criterion among three methods using (23). Note that no control algorithm is added in this case. Then, we put the Jacobian matrix estimator into the MPC control framework to observe the T_1 criterion under the same simulation environment. In addition, in the

pose control process, the computation time of each method is evaluated at each step, which is also regarded as a vital performance. Above three simulation evolutions for the pose motions executed by the soft manipulator are demonstrated in Fig. 3. It can be concluded that the analytical method owns the best estimation performance and make T_1 converge to 0 most quickly and smoothly in the control system, but the computation cost is at least ten times that of other methods. In LKF, there are frequent oscillations at the beginning and converge in the process of gradually decaying oscillations due to the inappropriate iteration direction, but this will take a longer time. The hybrid method has a more remarkable improvement in accuracy, faster and smoother convergence than LKF, and the computation cost is basically the same as LKF (except the initial point). In summary, the hybrid method can accurately and quickly estimate the Jacobian matrix to guide the pose motion of the soft manipulator.

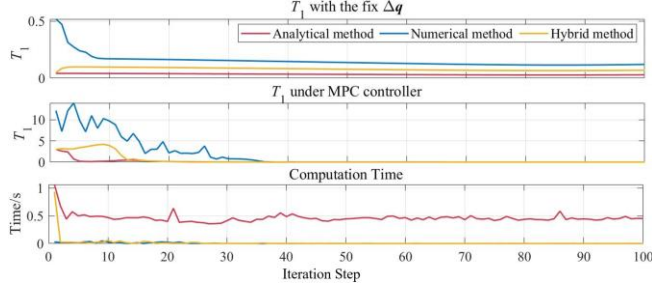


Fig. 3 Profiles of the T_1 and computation time among analytical, numerical and hybrid methods.

As mentioned in Section 3.3, there will be some kinematic singularities when the analytical method is applied. Consider this simulation case: we command the soft manipulator to move the desired pose $X^* = [200, 200, 50, 0, 0, 0]^T$ with the MPC controller. The evolution of position error e_p and q are shown in Fig.4. Obviously, it encounters a kinematic singularity in step 7, that is, the length of all q_{ij} is identical. At this time, the analytical method cannot estimate the Jacobian matrix, and the position error remains a constant that is not 0, so that the pose control failed.

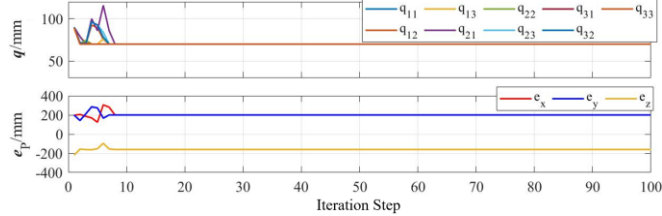


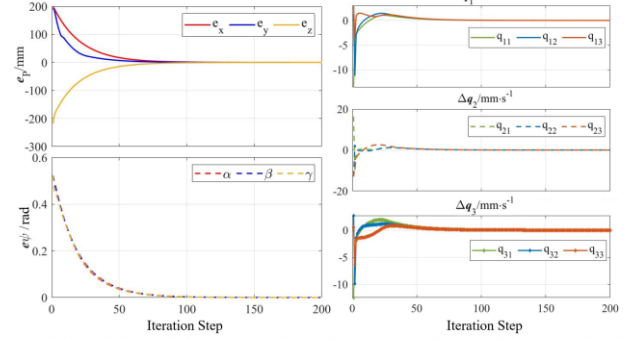
Fig. 4 Profiles of the joint vector and position error under the kinematic singularities.

B. MPC-based Pose Control

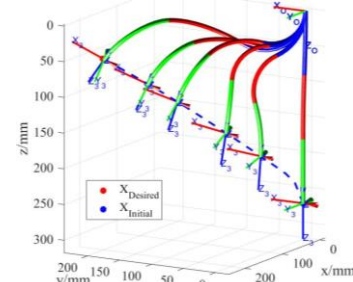
The effect of the proposed MPC controller for the pose control of the soft manipulator is verified and compared with other control algorithms to emphasize the superiority of ours. It is worth mentioning that all simulations in this section employ the hybrid method to construct the Jacobian matrix.

Consider the following simulation case: we command the soft manipulator to move the desired pose $X^* = [200, 200, 50, \pi/6, \pi/6, \pi/6]^T$ with the MPC controller. Fig.5 (a) shows the evolution of e_p , orientation error e_ϕ , and respective control input Δq . It can be seen that

under the action of the MPC controller, the error curve converges to 0 smoothly and quickly, and the joint speed is also kept small and smooth, which can ensure safety in actual operation tasks. To describe the motion process in more detail, Fig.7 demonstrates the 3D trajectory of the end-effector and the 3D shape deformation of the soft manipulator. The simulation video is publicly available at <https://github.com/zyyzyuibang/video/raw/main/video.mp4>.



(a) Profiles of the position error, orientation error and control input under the MPC controller with $G = 5$.



(b) 3D trajectory and the shape deformation
Fig. 5 Simulation results of the pose control based on MPC

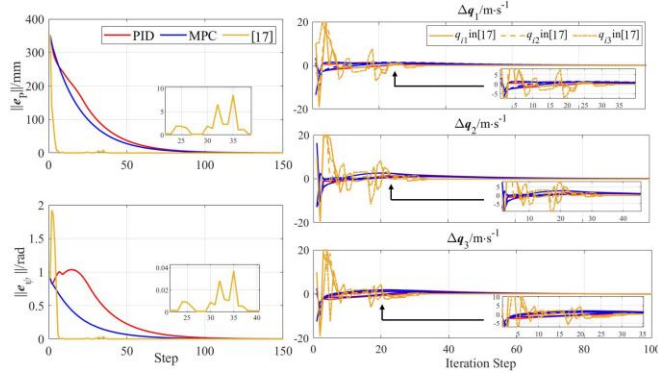
Finally, under the same conditions, the simulation experiment is carried out with the classical PID controller and an optimization controller for comparison. The recently studied optimization controller is stated in [17] as follows

$$\arg \min_{\Delta q} \lambda_1 \|J^v \Delta q - \Delta P\|_2^2 + \lambda_2 \|J^w \Delta q - \Delta \phi\|_2^2 + \lambda_3 \|\Delta q\|_2^2 \quad (36)$$

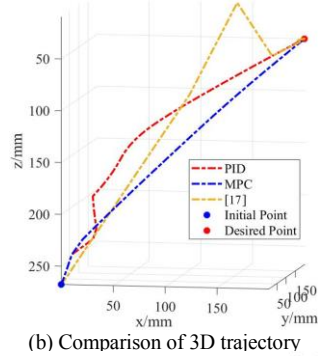
$$\text{s.t. } \Delta q \leq b \text{ and } q_{\min} \leq q \leq q_{\max}.$$

where $b = 20 \cdot \text{ones}(9, 1) \in \mathbb{R}^9$ to limit Δq . This method transforms the control problem into an optimization solution that satisfies the constraints and minimizes the cost function. According to the author's understanding, it needs to adjust the weight of λ_1 , λ_2 , and λ_3 to find the balance in the cost function. It seems hard to achieve the ideal minimization of all three items simultaneously. As shown in Fig.6(a), the optimization controller can achieve the fastest convergence of the position error, because the first item has the largest order of magnitude. While the second term is very small, in order to ensure orientation convergence, we set λ_2 to a larger number to increase the proportion of the second term, which means the stability of the third item is sacrificed. It can be observed that Δq oscillates violently and lasts for a long time, which is harmful to the prototype. In addition, this oscillation is likely to be amplified during the experimental process, eventually leading to control system divergence. Fig. 6(b) also displays the 3D trajectory for comparison. When we increase the weight of the third item to reduce Δq oscillation, the weight

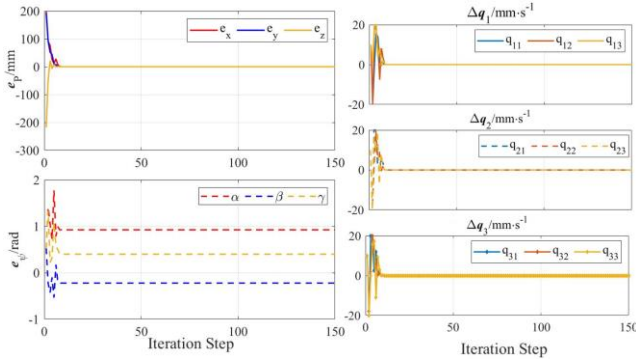
of the second item is relatively reduced, causing the failure of the orientation control (Fig. 6(c)). In conclusion, our proposed control framework achieves better performance relative to the overshoot, motion distance, and stability of control input. The videos of the PID and optimization controller are available at <https://github.com/zyyziubang/video/raw/main/PID.mp4> and <https://github.com/zyyziubang/video/raw/main/optimization%20controller.mp4>



(a) Profiles of the error and control input among three controllers.



(b) Comparison of 3D trajectory



(c) Profiles of the optimization controller with large λ_3 .

Fig. 6 Comparison of simulation results with three controllers.

VI. CONCLUSION

In this paper, A closed-loop control framework based on MPC is proposed to control the end position and orientation of the multi-section soft manipulator. PCC kinematics is introduced as the foundation. As for Jacobian matrix estimation, a novel hybrid Jacobian matrix estimation method is developed, which combines the advantages of analytical and numerical methods. It effectively avoids singularity, reduces the computation time, and also improves the convergence speed. The MPC controller with exponentially weighted is designed to control the pose of the soft manipulator. Simulations are conducted to verify the

feasibility and effectiveness of this work. Compared with the other controllers, it has a smaller overshoot, shorter motion distance and more stable dynamic process. In the future, we will verify our control framework through experiments.

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